LESSON 7 • TERM 3 • LEVEL 1

Fill the Grid

One of each, in every row and column

TODAY WE WILL

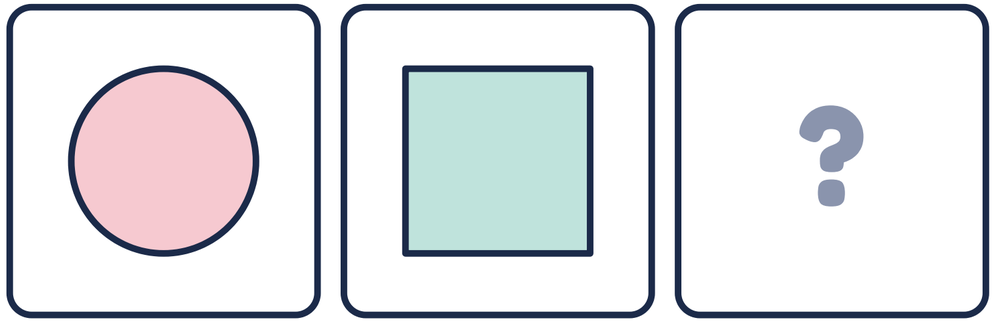
Fill a grid so every row and column has one of each.



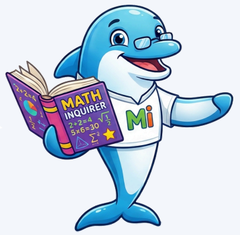
2–8 MIN

Warm-up: which is missing?

A row has one of each: a circle, a square and a triangle.



A circle and a square are here. Which shape is missing?



Look along the row!

Which digit is missing?

2–8 MIN

The digits 1, 2, 3 and 4 are used once each.

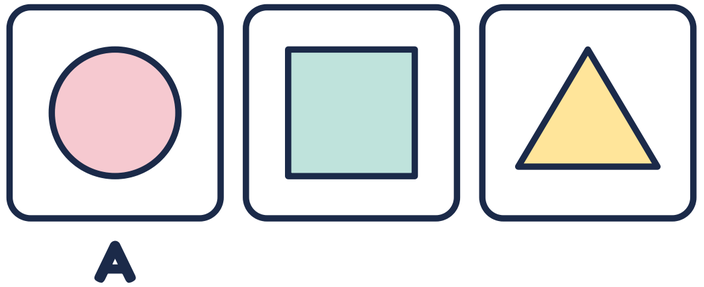
1	3	4	?

Which digit is missing?

Which row has one of each?

2–8 MIN

One of each means a circle, a square and a triangle, once each.



A



B

Which row has one of each?

Sneak-peek puzzles

8–12 MIN

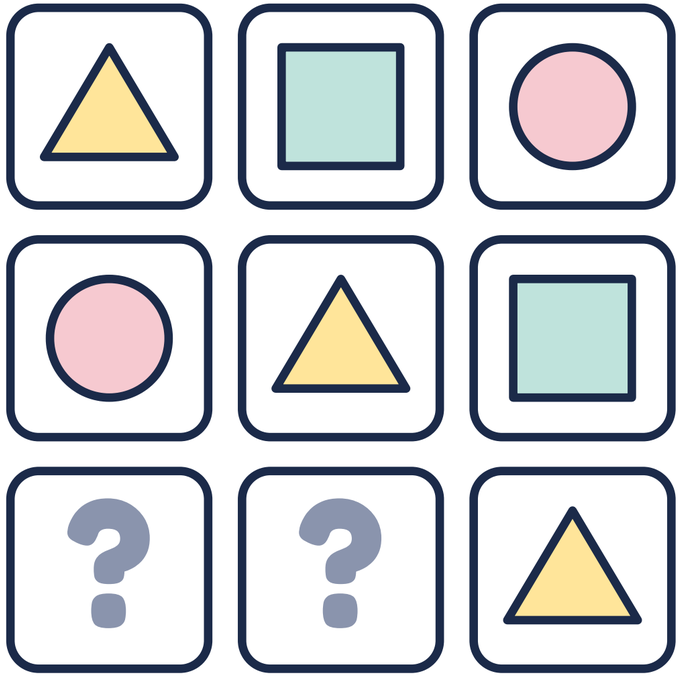
PUZZLE 1: anyone can try

Each row has one of each. Which shape goes in the empty box?



PUZZLE 2: think

Which shape goes in the first ? of the bottom row? I know because ___.



Explore: one of each in every row

12–23 MIN

1. Put mat 1 in front of you.
2. Take 3 circles, 3 squares and 3 triangles.
3. Fill the mat. Every row has a circle, a square and a triangle.

Rows only, for now.

Explore: one of each in every column

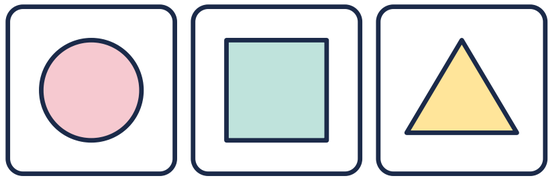
12–23 MIN

1. A column goes down.
2. Look down each column. Is there one of each?
3. Move tiles until every row and every column has one of each.

Take your time. Many ways are right.

Explore: the same top row

12–23 MIN



1. Keep this top row.
2. Fill the other two rows. Every row and every column has one of each.
3. How many different grids can you make?

Make one. Then make a different one.

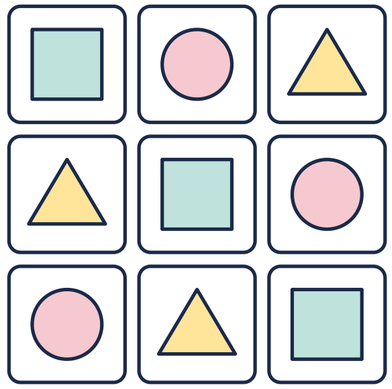
The rule

23–34 MIN

THE RULE

One of each in every row.

One of each in every column.



Look along the row. Look down the column.

23–34 MIN

ALONG THE ROW

What is missing?



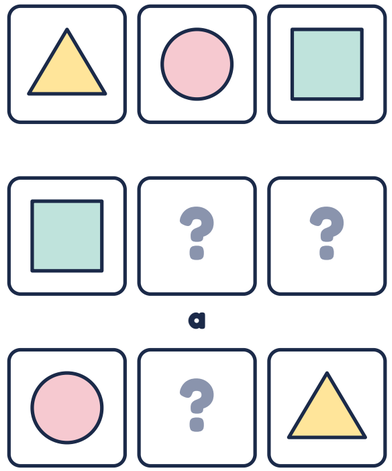
DOWN THE COLUMN

What is missing?

- ?
- circle
- triangle

Use the row and the column

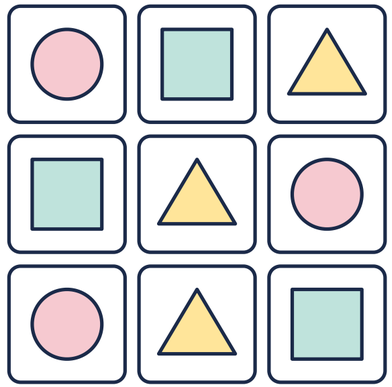
23–34 MIN



Look at box a. Its row has a square. Its column has a circle.

Is the grid finished?

23–34 MIN



Check every row. Check every column. Is it correct?

Your turn

23–34 MIN

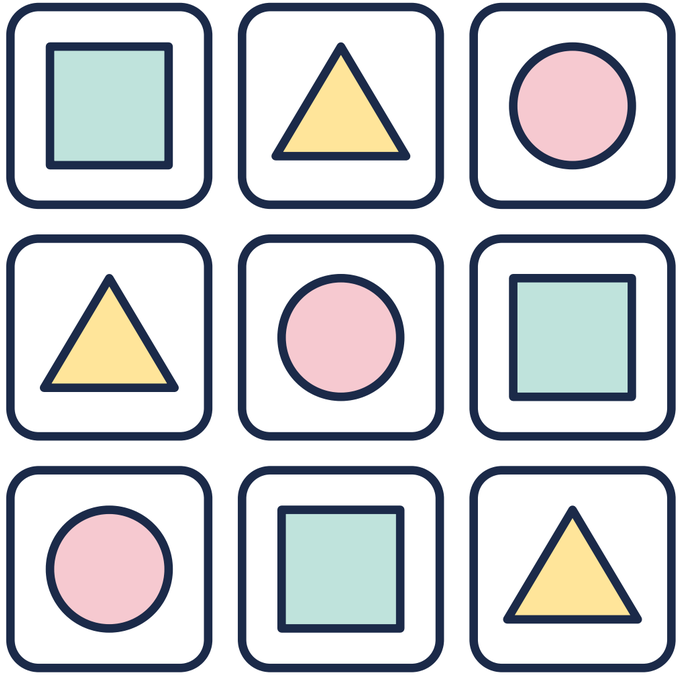
ROW

Which shape goes in the ? box?



GRID

Is the grid correct? Circle Yes or No.



Write both answers on your paper.

Check: your turn

23–34 MIN

Row: a triangle. Grid: Yes, it is correct.

34–38 MIN

Be the shapes

Stand up. I call a shape.

Make it with your arms or your body.

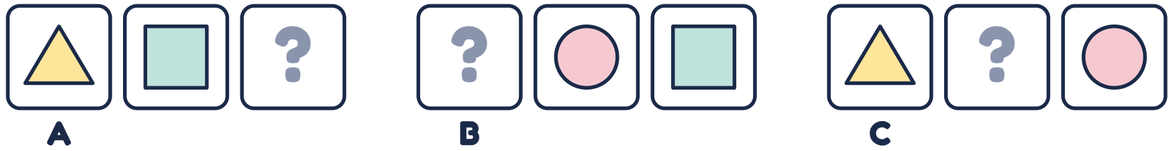
Sip some water.

Workbook page 1: Start here

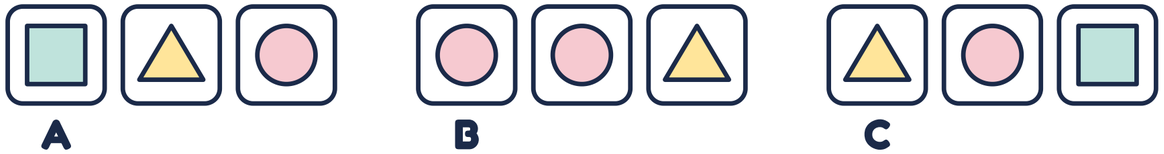
38–70 MIN

Questions 1 and 2. Take your time.

S1 Each row has a circle, a square and a triangle. Which shape goes in the box with the ? mark in rows A, B and C?



S2 Look at the three rows. Tick True or False for each statement.



38–70 MIN

Check page 1

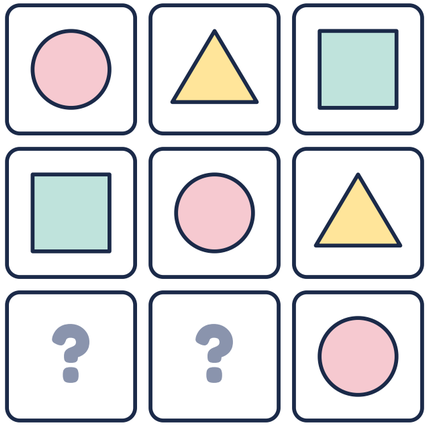
S1	circle; triangle; square
S2	A true; B false; C true

Workbook page 2: Start here

38–70 MIN

Questions 3 to 5. Take your time.

S3 Every row and every column has a circle, a square and a triangle. Look at the bottom left box.



S4 A child fills a grid. Row 1 is circle, square, triangle. Row 2 is square, triangle, circle. Row 3 is circle, square, triangle. Every row has one of each. Is the grid correct?

S5 Every row of a 3 by 3 grid has one circle, and every column has one circle. How many circles are in the whole grid?

38–70 MIN

Check page 2

S3	triangle; the column has a circle and a square
S4	No
S5	3

38–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	Two clues, how many shapes, what stays correct
Go further (page 4)	Four by Four, and bigger grids
Investigation (page 5)	Grid builder
Check	Finish and check your own work

Everyone may choose. Not finished? Pages 3 to 5 are for home too.

Talk and explain

70–76 MIN

A CHILD SAYS

I can fill every empty box by looking along the row. I never need the column.

Is the child right? Test it with your tiles.

I think … because …

Back to the sneak-peek

70–76 MIN

Puzzle 1: circle. Puzzle 2: a square. The column has a triangle and a circle, and the row alone could not tell us.

Exit ticket

76–82 MIN

On your own. No looking back.

E1 A row has a circle and a square. Which shape is missing from the row? Circle one.

E2 Which row has one of each? Choose the correct answer. Write its number in the bracket.

E3 A box has a circle in its row and a square in its column. Which shape goes in the box? Circle one.

Check the exit ticket

76–82 MIN

E1	triangle
E2	(2) square, triangle, circle
E3	triangle

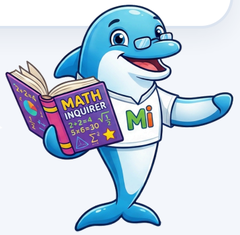
Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ finding the missing shape ☐ using the row and the column ☐ checking the whole grid ☐ something else
What helped me	☐ my tiles ☐ looking along and down ☐ checking my answer ☐ I was not stuck
Next time I will try	☐ using my tiles ☐ checking every column ☐ checking my answer ☐ asking a question
	☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Take your time. You can pass.



85–90 MIN

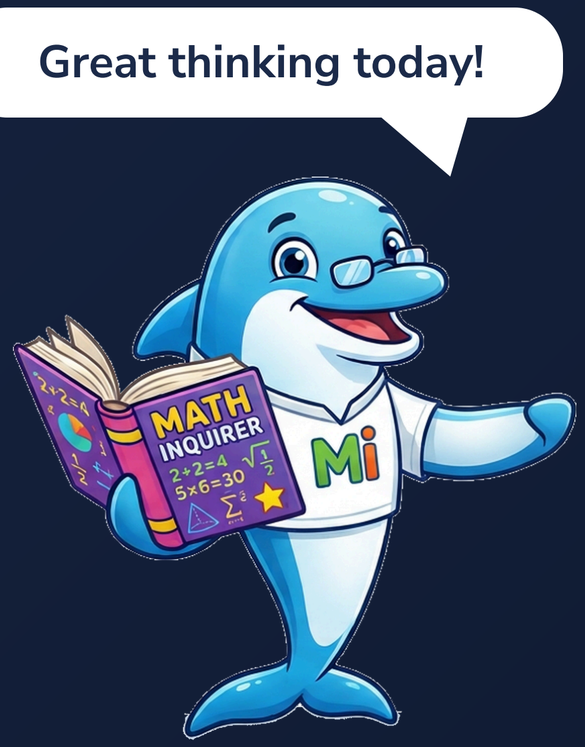
Well done, inquirers!

Show a grown-up one grid you filled today.

NEXT LESSON

Make a Table

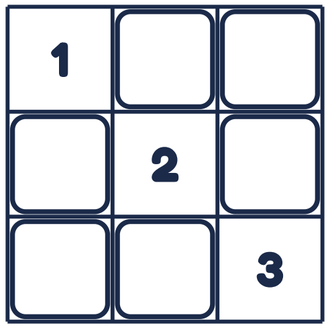
Have ready: the strip page, cut out by a grown-up (14 strips).



OPTIONAL

Go further: three in a row

Mei has a 3 by 3 grid with the digits 1, 2 and 3.



Every row and every column has 1, 2 and 3 once. Can you fill it?

OPTIONAL

If we have time: pick one

Activity	What we do
Is it always true?	Two circles can never be next to each other. Test it.
Make a puzzle	Cover three boxes. Ask a grown-up to find them.
Teach it back	Show a toy how to find a missing box.

Optional. Everyone may join.